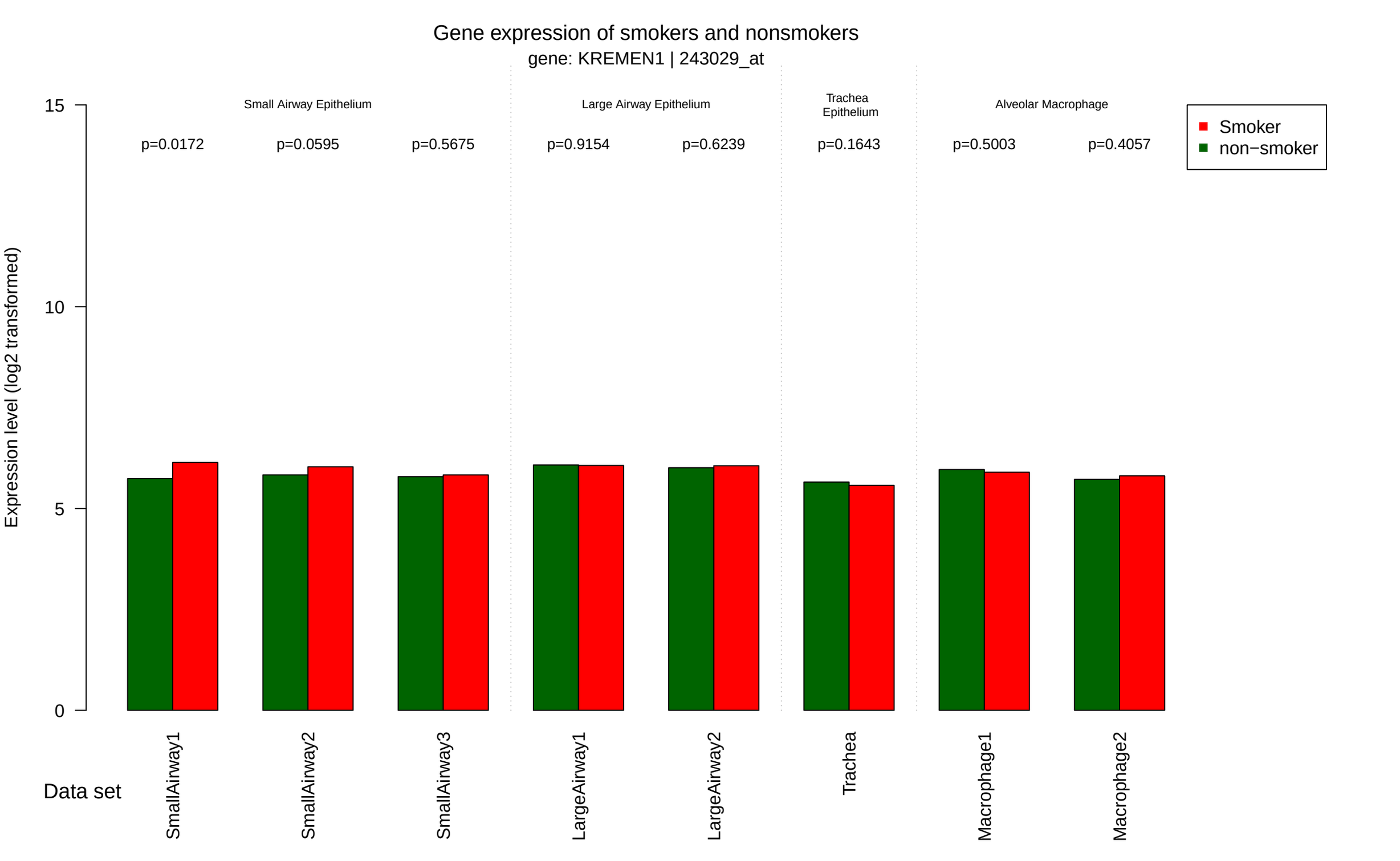
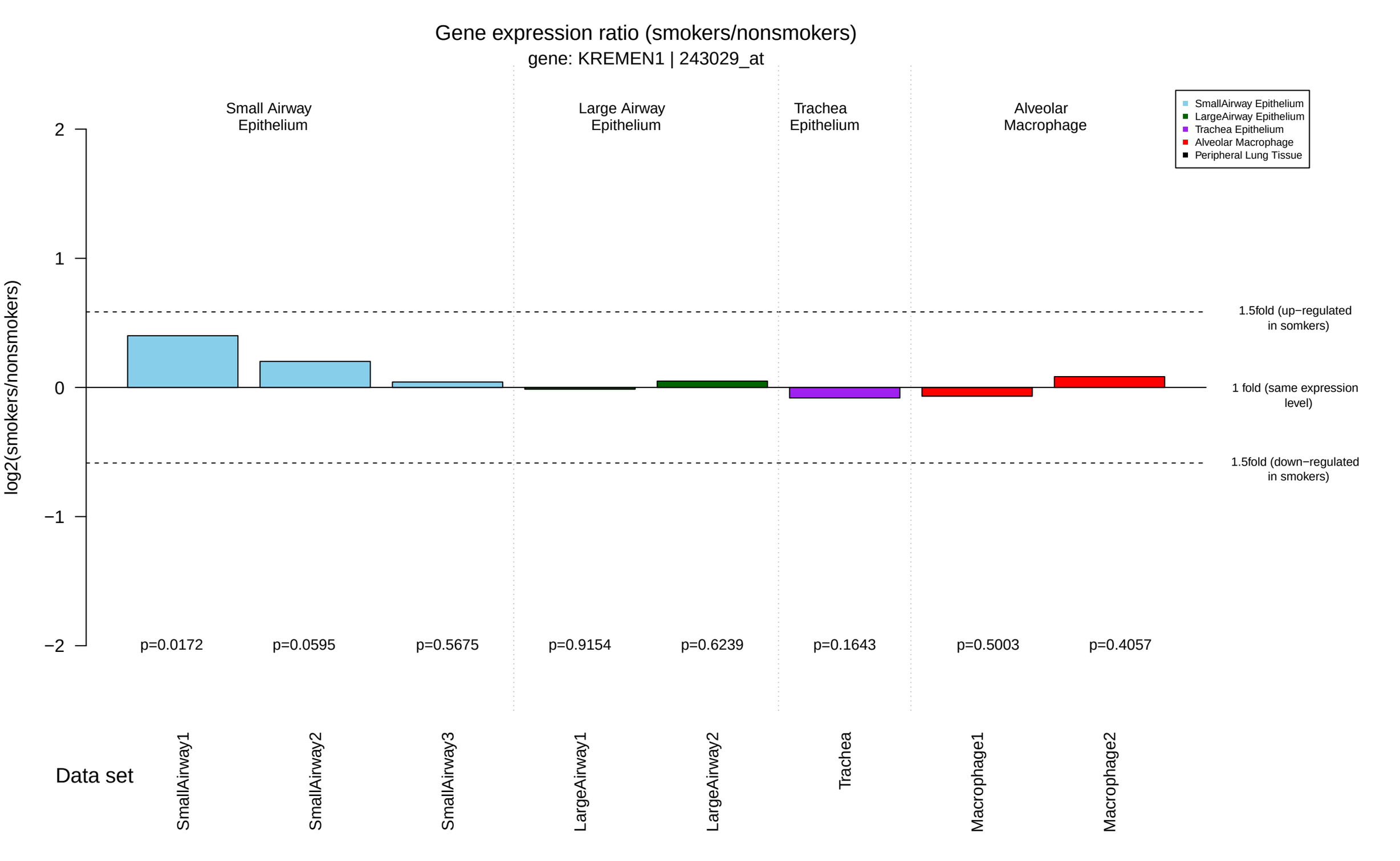
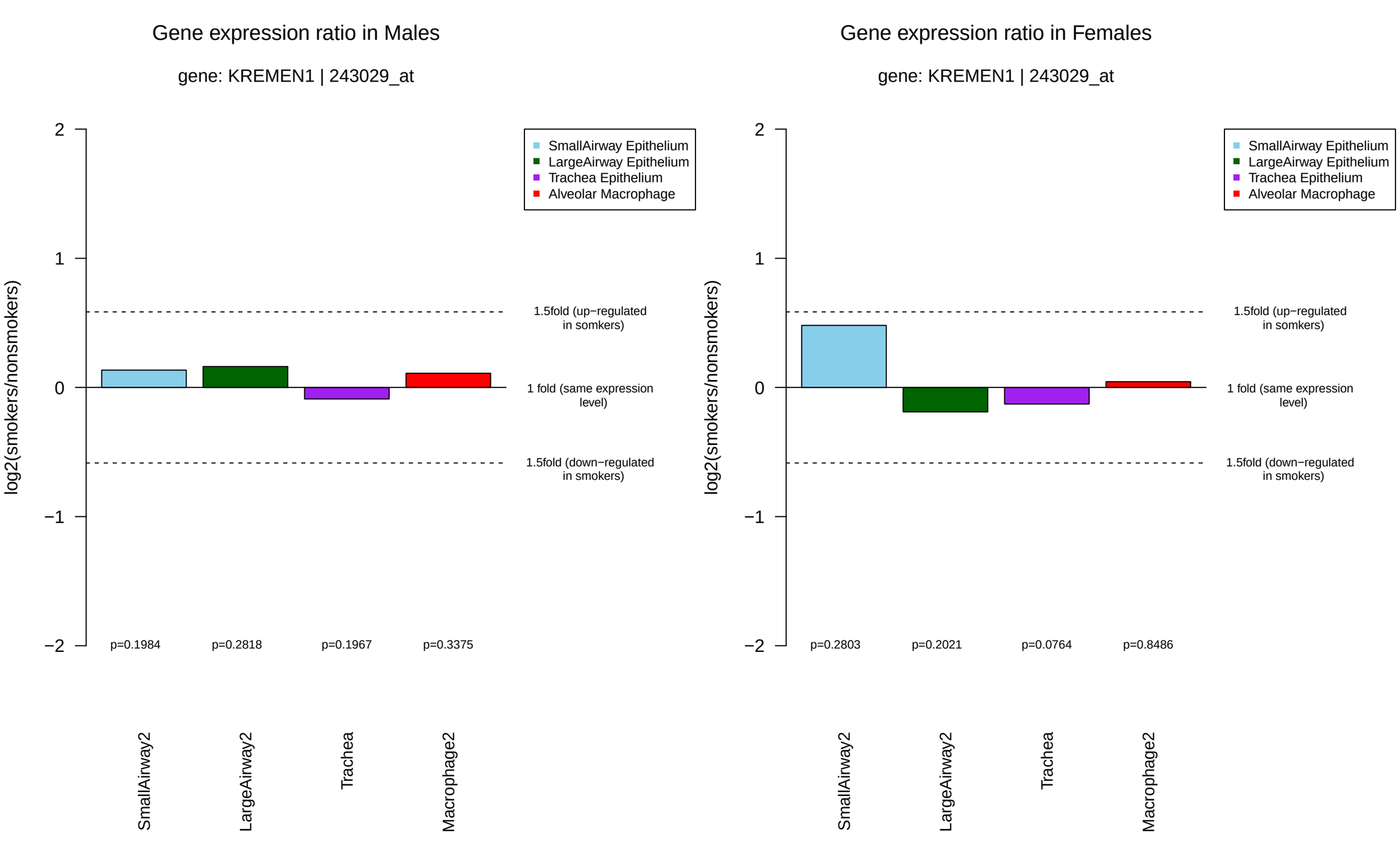
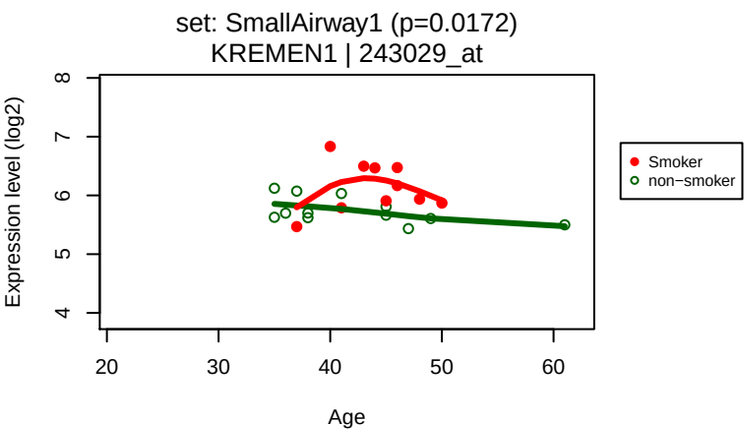
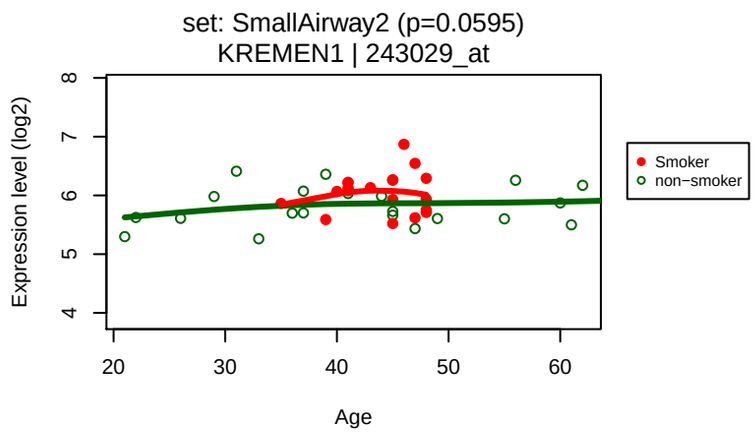
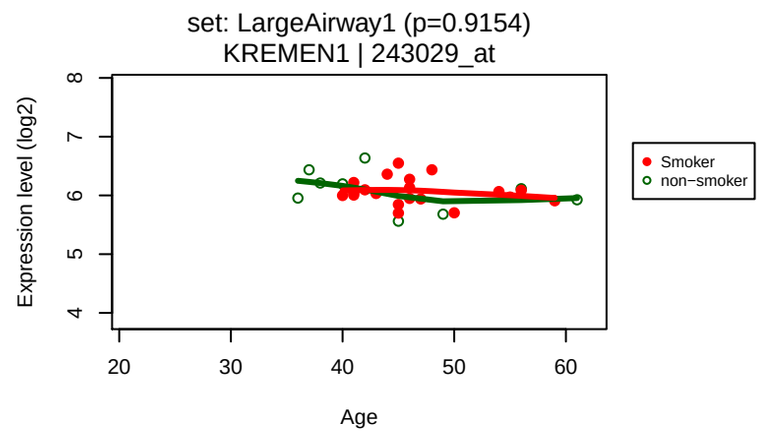

Gene expression of smokers and nonsmokers
gene: KREMEN1 | 243029_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0172
p=0.0595
p=0.5675
p=0.9154
p=0.6239
p=0.1643
p=0.5003
p=0.4057
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

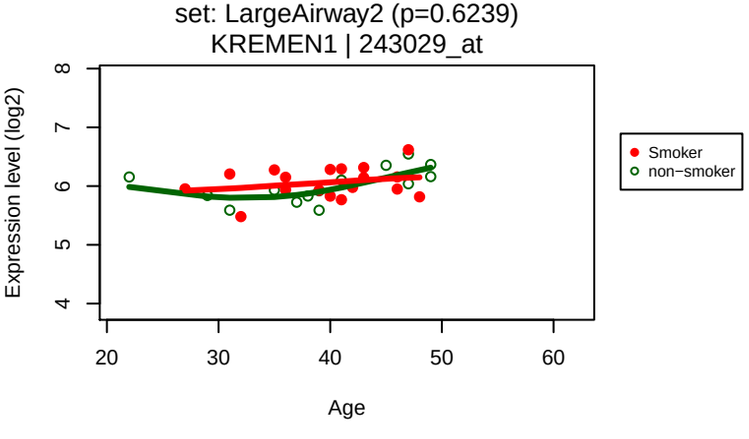

Gene expression ratio (smokers/nonsmokers)
gene: KREMEN1 | 243029_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0172
p=0.0595
p=0.5675
p=0.9154
p=0.6239
p=0.1643
p=0.5003
p=0.4057
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

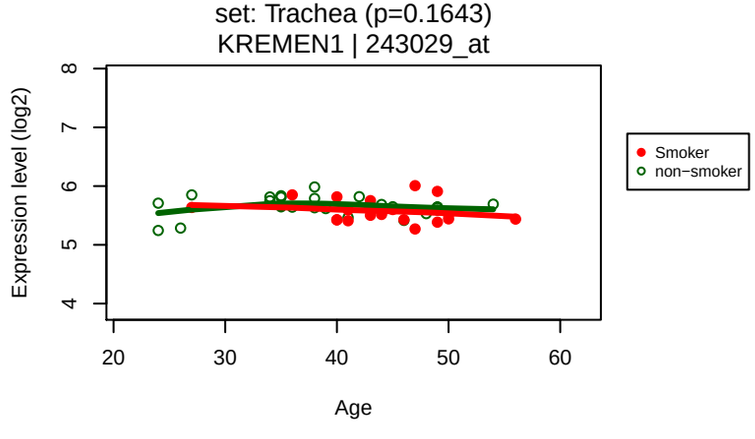

Gene expression ratio in Males
gene: KREMEN1 | 243029_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1984
p=0.2818
p=0.1967
p=0.3375
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KREMEN1 | 243029_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2803
p=0.2021
p=0.0764
p=0.8486
SmallAirway2
LargeAirway2
Trachea
Macrophage2

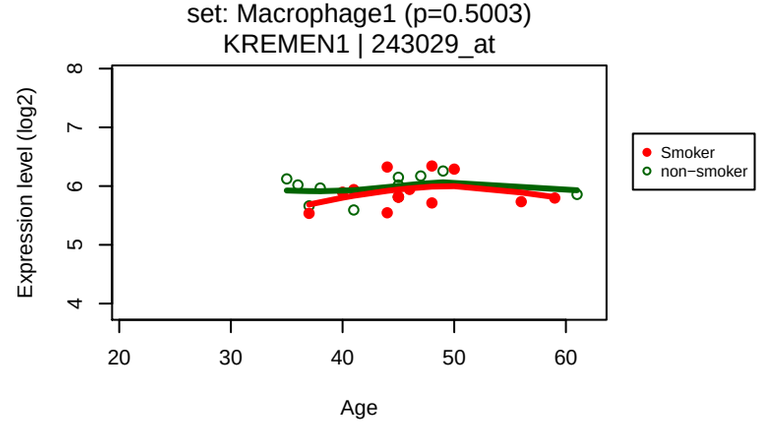

set: SmallAirway1 (p=0.0172)
KREMEN1 | 243029_at
Expression level (log2)
Age
Smoker
non−smoker

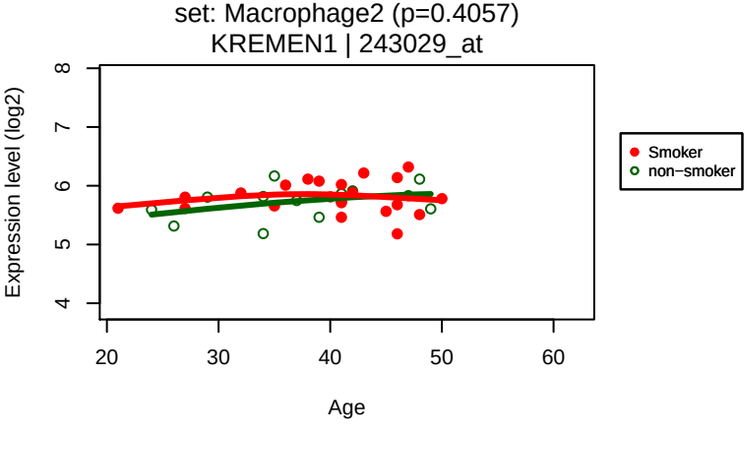

set: SmallAirway2 (p=0.0595)
KREMEN1 | 243029_at
Expression level (log2)
Age
Smoker
non−smoker

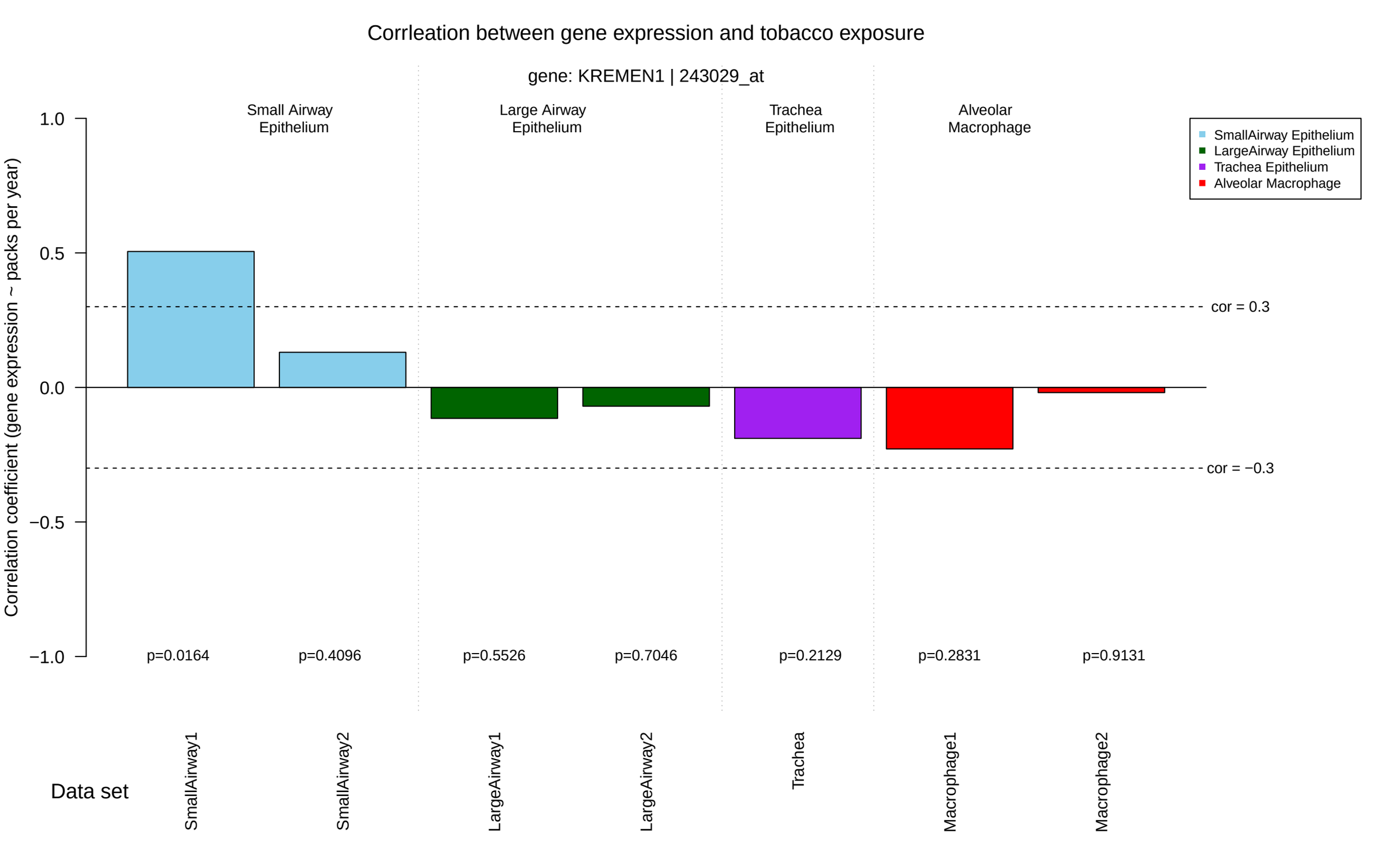

set: LargeAirway1 (p=0.9154)
KREMEN1 | 243029_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6239)
KREMEN1 | 243029_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1643)
KREMEN1 | 243029_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5003)
KREMEN1 | 243029_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4057)
KREMEN1 | 243029_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KREMEN1 | 243029_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0164
p=0.4096
p=0.5526
p=0.7046
p=0.2129
p=0.2831
p=0.9131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2